Chapter 6 Jointly Distributed Random Variables

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6.1 Joint Distribution Functions

Thus far, we have concerned ourselves only with probability distributions for single random variables. However, we are often interested in probability statements concerning two or more random variables. In order to deal with such probabilities, we define, for any two random variables X and Y, the *joint cumulative probability distribution function* of X and Y by

$$F(a,b)=P\{X\le a,Y\le b\} \quad -\infty<a,b<\infty$$

All joint probability statements about X and Y can, in theory, be answered in terms of their joint distribution function. For instance,

$$P(a1<X\le a2,\ b1<Y\le b2)=F(a2,b2)+F(a1,b1)-F(a1,b2)-F(a2,b1) \tag{1.1}$$

whenever a1<a2, b1<b2. To verify **Equation (1.1)**, note that for a1<a2,

$$P(X\le a2,\ Y\le b)=P(X\le a1,\ Y\le b)+P(a1<X\le a2,\ Y\le b)$$

giving that

$$P(a1<X\le a2,\ Y\le b)=F(a2,b)-F(a1,b) \tag{1.2}$$

Also, because for b1<b2,

$$P(a1<X\le a2,\ Y\le b2)=P(a1<X\le a2,\ Y\le b1)+P(a1<X\le a2,\ b1<Y\le b2)$$

we have that when a1<a2, b1<b2

$$P(\ a\ 1<X\le a\ 2\ ,\ \ b\ 1<Y\le b\ 2\ )=P(\ a\ 1<X\le a\ 2\ ,\ Y\le b\ 2\ )-P(\ a\ 1<X\le a\ 2\ ,\ Y\le b\ 1\ )=F(\ a\ 2\ ,\ b\ 2\ )-F(\ a\ 1\ ,\ b\ 2\ )-F(\ a\ 2\ ,\ b\ 1\ )+F(\ a\ 1\ ,\ b\ 1\ )$$

where the final equality used **Equation (1.2)**.

When X and Y are discrete random variables, with X taking on one of the values xi, i≥1, and Y one of the values yj, j≥1, it is convenient to define the *joint probability mass function* of X and Y by

$$p(x,y)=P(X=x,Y=y)$$

Using that the event {X=x} is the union of the mutually exclusive events {X=x,Y=yj}, j≥1, it follows that the probability mass function of X can be obtained from the joint probability mass function by

$$p_X(x)=P(X=x)=P(\cup_j\{X=x,Y=y_j\})=\sum_j P(X=x,Y=y_j)=\sum_j p(x,y_j)$$

Similarly, the probability mass function of Y is obtained from

$$p_Y(y)=\sum_i p(x_i,y)$$

Example 1a

Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. If we let X and Y denote, respectively, the number of red and white balls chosen, then the joint probability mass function of X and Y,p(i,j)=P{X=i,Y=j}, is obtained by noting that X=i, Y=j if, of the 3 balls selected, i are red, j are white, and 3−i−j are blue. Because all subsets of size 3 are equally likely to be chosen, it

follows that

$$p(i,j) = \binom{3}{i}\binom{4}{j}\binom{5}{3-i-j}\Big/\binom{12}{3}$$

Consequently,

$$p(0,0) = \binom{5}{3}\Big/\binom{12}{3} = \frac{10}{220} \quad p(0,1) = \binom{4}{1}\binom{5}{2}\Big/\binom{12}{3} = \frac{40}{220} \quad p(0,2) = \binom{4}{2}\binom{5}{1}\Big/\binom{12}{3} = \frac{30}{220}$$
$$p(0,3) = \binom{4}{3}\Big/\binom{12}{3} = \frac{4}{220} \quad p(1,0) = \binom{3}{1}\binom{5}{2}\Big/\binom{12}{3} = \frac{30}{220} \quad p(1,1) = \binom{3}{1}\binom{4}{1}\binom{5}{1}\Big/\binom{12}{3} = \frac{60}{220}$$
$$p(1,2) = \binom{3}{1}\binom{4}{2}\Big/\binom{12}{3} = \frac{18}{220} \quad p(2,0) = \binom{3}{2}\binom{5}{1}\Big/\binom{12}{3} = \frac{15}{220} \quad p(2,1) = \binom{3}{2}\binom{4}{1}\Big/\binom{12}{3} = \frac{12}{220}$$
$$p(3,0) = \binom{3}{3}\Big/\binom{12}{3} = \frac{1}{220}$$

These probabilities can most easily be expressed in tabular form, as in **Table 6.1**. The reader should note that the probability mass function of X is obtained by computing the row sums, whereas the probability mass function of Y is obtained by computing the column sums. Because the individual probability mass functions of X and Y thus appear in the margin of such a table, they are often referred to as the *marginal probability mass functions* of X and Y, respectively.

Table 6.1 P{X=i,Y=j}.

i \ j	0	1	2	3	Row sum $= P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum $= P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

Example 1b

Suppose that 15 percent of the families in a certain community have no children, 20 percent have 1 child, 35 percent have 2 children, and 30 percent have 3. Suppose further that in each family each child is equally likely (independently) to be a boy or a girl. If a family is chosen at random from this community, then B, the number of boys, and G, the number of girls, in this family will have the joint probability mass function shown in **Table 6.2**.

Table 6.2 P{B=i,G=j}.

i \ j	0	1	2	3	Row sum $= P\{B = i\}$
0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	0	.2000
3	.0375	0	0	0	.0375
Column sum $= P\{G = j\}$	.3750	.3875	.2000	.0375	

The probabilities shown in **Table 6.2** are obtained as follows:

P{ B=0,G=0 } =P{ no children }=.15 P{ B=0,G=1 } =P{ 1 girl and total of 1 child } =P{ 1 child }P{ 1 girl | 1 child }=(.20)(1 2) P{ B=0,G=2 } =P{ 2 girls and total of 2 children } =P{ 2 children }P{ 2 girls | 2 children }=(.35) (1 2) 2

We leave the verification of the remaining probabilities in the table to the reader.

Example 1c

Consider independent trials where each trial is a success with probability p. Let Xr denote the number of trials until there have been r successes, and let Ys denote the number of trials until there have been s failures. Suppose we want to derive their joint probability mass function P(Xr=i,Ys=j). To do so, first consider the case i<j. In this case, write

P(Xr=i,Ys=j)=P(Xr=i)P(Ys=j|Xr=i)

Now, if there have been r successes after trial i then there have been i−r failures by that point. Hence, the conditional distribution of Ys, given that Xr=i, is the distribution of i plus the number of additional trials after trial i until there have been an additional s−i+r failures. Hence,

P(Xr=i,Ys=j)=P(Xr=i)P(Ys−i+r=j−i) , i<j

Because Xr is a negative binomial random variable with parameters (r,p) and Ys−i+r is a negative binomial random variable with parameters (s−i+r,1−p), the preceding yields

P(X r =i, Y s =j)=(i−1 r−1) p r (1−p) i−r (j−i−1 s−i+r−1) (1−p) s−i+r p j−s−r , i<j

We leave it as an exercise to determine the analogous expression when j<i.

We say that X and Y are *jointly continuous* if there exists a function f(x,y), defined for all real x and y, having the property that for every set C of pairs of real numbers (that is, C is a set in the two-dimensional plane),

$$P\{(X,Y)\in C\}= \iint_{(x,y)\in C} f(x,y)\,dx\,dy \qquad (1.3)$$

The function f(x,y) is called the *joint probability density function* of X and Y. If A and B are any sets of real numbers, then by defining C={(x,y):x∈A,y∈B}, we see from **Equation (1.3)** that

$$P\{ X\in A,Y\in B \}= \int B \int A f(x,y)\, dx\, dy \tag{1.4}$$

Because

$$F(a,b) =P\{ X\in(-\infty, a] ,Y\in(-\infty,b] \} = \int -\infty\ b \int -\infty\ a f(x,y)\, dx\, dy$$

it follows, upon differentiation, that

$$f(a,b)= \partial\ 2\ \partial a \partial b\ F(a,b)$$

wherever the partial derivatives are defined. Another interpretation of the joint density function, obtained from **Equation (1.4)**, is

$$P\{a<X<a+da,b<Y<b+db\} = \int b\ d+db \int a\ a+da f(x,y)\, dx\, dy \approx f(a,b)\, da\, db$$

when da and db are small and f(x,y) is continuous at a, b. Hence, f(a,b) is a measure of how likely it is that the random vector (X, Y) will be near (a, b).

If X and Y are jointly continuous, they are individually continuous, and their probability density functions can be obtained as follows:

$$P\{ X\in A \} =P\{ X\in A,Y\in(-\infty,\infty) \} = \int A \int -\infty\ \infty f(x,y)\, dy\, dx = \int A f\ X\ (x)\, dx$$

where

$$f\ X\ (x)= \int -\infty\ \infty f(x,y)\, dy$$

is thus the probability density function of X. Similarly, the probability density function of Y is given by

$$f\ Y\ (y)= \int -\infty\ \infty f(x,y)\, dx$$

Example 1d

The joint density function of X and Y is given by

$$f(x,y)=\{\ 2\ e\ -x\ e\ -2y \quad 0<x<\infty,\ 0<y<\infty\ 0 \quad \text{otherwise}$$

Compute (a) P{X>1,Y<1}, (b) P{X<Y}, and (c) P{X<a}.

Solution

a. $P(X>1, Y<1)= \int 0\ 1 \int 1\ \infty\ 2\ e\ -x\ e\ -2y\ dx\, dy$

   Now,
   $\int 1\ \infty\ e\ -x\, dx=-\ e\ -x\ |\ 1\ \infty = e\ -1$

   giving that
   $P(X>1, Y<1)= e\ -1 \int 0\ 1\ 2\ e\ -2y\, dy= e\ -1\ (1-\ e\ -2\ )$

b. $P\{X<Y\} = \iint (\ x,y\ ):x<y\ 2\ e\ -x\ e\ -2y\ dx\, dy = \int 0\ \infty \int 0\ y\ 2\ e\ -x\ e\ -2y\ dx\, dy = \int 0\ \infty\ 2\ e\ -2y\ (1-\ e\ -y\ )dy = \int 0\ \infty\ 2\ e\ -2y\, dy- \int 0\ \infty\ 2\ e\ -3y\, dy =1-\ 2\ 3 = 1\ 3$

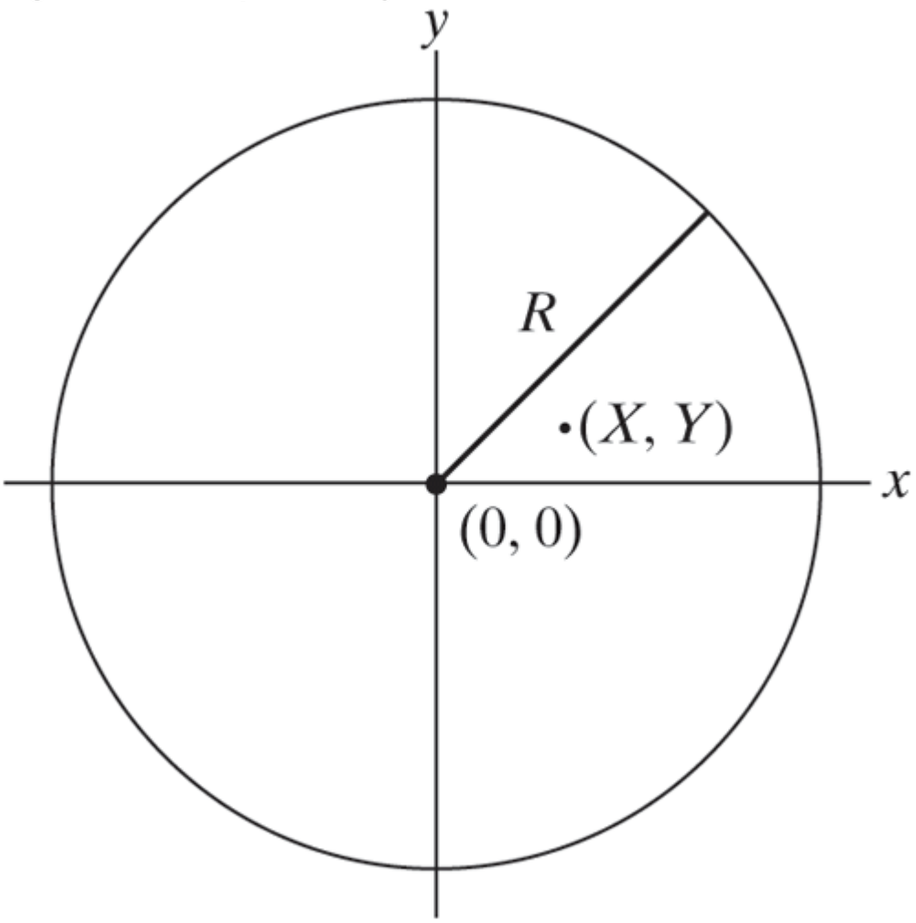
c. $P\{X<a\} = \int_0^a \int_0^\infty 2e^{-2y} e^{-x}\, dy\, dx = \int_0^a e^{-x}\, dx = 1 - e^{-a}$

Example 1e

Consider a circle of radius R, and suppose that a point within the circle is randomly chosen in such a manner that all regions within the circle of equal area are equally likely to contain the point. (In other words, the point is uniformly distributed within the circle.) If we let the center of the circle denote the origin and define X and Y to be the coordinates of the point chosen (**Figure 6.1**), then, since (X, Y) is equally likely to be near each point in the circle, it follows that the joint density function of X and Y is given by

$$f(x,y) = \begin{cases} c & \text{if } x^2+y^2 \le R^2 \\ 0 & \text{if } x^2+y^2 > R^2 \end{cases}$$

Figure 6.1 Joint probability distribution.



for some value of c.

a. Determine c.
b. Find the marginal density functions of X and Y.
c. Compute the probability that D, the distance from the origin of the point selected, is less than or equal to a.
d. Find E [D].

Solution

a. Because
$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dy\, dx = 1$

it follows that
c $\iint x^2 + y^2 \le R^2 \, dy\, dx = 1$

We can evaluate $\int \int x^2 + y^2 \le R^2 \, dy\, dx$ either by using polar coordinates or, more simply, by noting that it represents the area of the circle and is thus equal to πR^2. Hence,
$c = 1\pi R^2$

b. $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\, dy = 1\pi R^2 \int x^2 + y^2 \le R^2 \, dy = 1\pi R^2 \int_{-a}^{a} dy$, where $a = R^2 - x^2$
$= 2\pi R^2 R^2 - x^2, \quad x^2 \le R^2$

and it equals 0 when $x^2 > R^2$. By symmetry, the marginal density of Y is given by
$f_Y(y) = 2\pi R^2 R^2 - y^2, \; y^2 \le R^2 = 0 \; y^2 > R^2$

c. The distribution function of $D = X^2 + Y^2$, the distance from the origin, is obtained as follows: For $0 \le a \le R$,
$F_D(a) = P\{X^2 + Y^2 \le a\} = P\{X^2 + Y^2 \le a^2\} = \iint x^2 + y^2 \le a^2 f(x,y)\, dy\, dx = 1\pi R^2 \iint x^2 + y^2 \le a^2 \, dy\, dx = \pi a^2 \pi R^2 = a^2 R^2$

where we have used the fact that $\int \int x^2 + y^2 \le a^2 \, dy\, dx$ is the area of a circle of radius *a* and thus is equal to $\pi\alpha^2$.

d. From part (c), the density function of D is
$f_D(a) = 2aR^2 \quad 0 \le a \le R$

Hence,
$E[D] = 2R^2 \int_0^R a^2 da = 2R^3$

Example 1f

The joint density of X and Y is given by

$f(x,y) = \{ e^{-(x+y)} \; 0 < x < \infty, \; 0 < y < \infty \; 0$ otherwise

Find the density function of the random variable X/Y.

Solution

We start by computing the distribution function of X/Y. For $a > 0$,

$F_{X/Y}(a) = P\{X Y \le a\} = \iint x/y \le a \; e^{-(x+y)} \, dx\, dy = \int_0^{\infty} \int_0^{ay} e^{-(x+y)} \, dx\, dy = \int_0^{\infty} (1 - e^{-ay}) e^{-y} dy = \{ -e^{-y} + e^{-(a+1)y}/(a+1) \} \mid_0^{\infty} = 1 - 1/(a+1)$

Differentiation shows that the density function of X/Y is given by $f_{X/Y}(a) = 1/(a+1)^2, 0 < a < \infty$.

We can also define joint probability distributions for n random variables in exactly the same manner as we did for n=2. For instance, the joint cumulative probability distribution function $F(a_1, a_2, \ldots, a_n)$ of the n random variables $X_1, X_2, \ldots, X_n$ is defined by

$F(a_1, a_2, \ldots, a_n) = P\{X_1 \le a_1, X_2 \le a_2, \ldots, X_n \le a_n\}$

Further, the n random variables are said to be *jointly continuous* if there exists a function $f(x_1, x_2, \ldots, x_n)$, called the *joint probability density function*, such that, for any set C in n-space,

$P\{(X_1, X_2, \ldots, X_n) \in C\} = \iint \cdots \int_{(x_1, \ldots, x_n) \in C} f(x_1, \ldots, x_n) dx_1\, dx_2 \cdots dx_n$

In particular, for any *n* sets of real numbers A1,A2,… ,An,

$$P\{X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n\} = \int_{A_n} \int_{A_{n-1}} \cdots \int_{A_1} f(x_1, \ldots, x_n)\, dx_1\, dx_2 \cdots dx_n$$

Example 1g The Multinomial Distribution

One of the most important joint distributions is the multinomial distribution, which arises when a sequence of n independent and identical experiments is performed. Suppose that each experiment can result in any one of r possible outcomes, with respective probabilities p1,p2,… ,pr,∑i=1rpi=1. If we let Xi denote the number of the n experiments that result in outcome number i, then

$$P\{X_1 = n_1, X_2 = n_2, \ldots, X_r = n_r\} = \frac{n!}{n_1!\, n_2! \cdots n_r!} p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r} \qquad (1.5)$$

whenever ∑i=1rni=n.

Equation (1.5) is verified by noting that any sequence of outcomes for the n experiments that leads to outcome i occurring ni times for i=1,2,… ,r will, by the assumed independence of experiments, have probability p 1 n 1 p 2 n2 … p r nr of occurring. Because there are n!/(n1!n2!… nr!) such sequences of outcomes (there are n!/n1! … nr! different permutations of n things of which n1 are alike, n2 are alike, … ,nr are alike), **Equation (1.5)** is established. The joint distribution whose joint probability mass function is specified by **Equation (1.5)** is called the *multinomial distribution*. Note that when r=2, the multinomial reduces to the binomial distribution.

Note also that any sum of a fixed set of the X i ' s will have a binomial distribution. That is, if N⊂{1,2,… ,r}, then ∑ i∈N X i will be a binomial random variable with parameters n and p= ∑ i∈N p i . This follows because ∑ i∈N X i represents the number of the n experiments whose outcome is in N, and each experiment will independently have such an outcome with probability ∑i∈Npi.

As an application of the multinomial distribution, suppose that a fair die is rolled 9 times. The probability that 1 appears three times, 2 and 3 twice each, 4 and 5 once each, and 6 not at all is

9!3!2!2!1!1!0!(16)3(16)2(16)2(16)1(16)1(16)0=9!3!2!2!(16)9

We can also use the multinomial distribution to analyze a variation of the classical birthday problem which asks for the probability that no 3 people in a group of size n have the same birthday when the birthdays of the n people are independent and each birthday is equally likely to be any of the 365 days of the year. Because this probability is 0 when n>730 (why is this), we will suppose that n≤730. To find the desired probability, note that there will be no set of 3 people having the same birthday if each of the 365 days of the year is the birthday of at most 2 persons. Now, this will be the case if for some i≤n/2 the event Ai occurs, where Ai is the event that the 365 days of the year can be partitioned into three groups of respective sizes i,n−2i, and 365−n+i such that every day in the first group is the birthday of exactly 2 of the n individuals, every day in the second group is the birthday of exactly 1 of the n individuals, and every day in the third group is the birthday of none of the n individuals. Now, because each day of the year is equally likely to be the birthday of an individual, it follows, for a given partition of the 365 days into three groups of respective sizes i, n−2i, and 365−n+i, that the probability each day in the first group is the birthday of exactly 2 of the n individuals, each day in the second group is the birthday of exactly 1 of the n individuals, and each day in the third group is the birthday of none of the n individuals is equal to the multinomial probability

n!(2!)i(1!)n−2i(0!)365−n+i(1365)n.

As the number of partitions of the 365 days of the year into 3 groups of respective sizes i,n−2i,365−n+i is 365! i!(n−2i)!(365−n+i)! , it follows that

$$P(A_i)=\frac{365!}{i!(n-2i)!(365-n+i)!}\frac{n!}{2^i(1365)^n}, \quad i\le n/2$$

As the events A_i, $i\le n/2$, are mutually exclusive we have that

$$P\{\text{no set of three with same birthday}\}=\sum_{i=0}^{[n/2]}\frac{365!}{i!(n-2i)!(365-n+i)!}\frac{n!}{2^i(1365)^n}$$

When $n=88$, the preceding gives

$$P\{\text{no set of three with same birthday}\}=\sum_{i=0}^{44}\frac{365!}{i!(88-2i)!(277+i)!}\frac{88!}{2^i(1365)^{88}}\approx .504$$

6.2 Independent Random Variables

The random variables X and Y are said to be *independent* if, for any two sets of real numbers A and B,

$$P\{X\in A, Y\in B\}=P\{X\in A\}P\{Y\in B\} \tag{2.1}$$

In other words, X and Y are independent if, for all A and B, the events $E_A=\{X\in A\}$ and $F_B=\{Y\in B\}$ are independent.

It can be shown by using the three axioms of probability that **Equation (2.1)** will follow if and only if, for all a, b

$$P\{X\le a, Y\le b\}=P\{X\le a\}P\{Y\le b\}$$

Hence, in terms of the joint distribution function F of X and Y, X and Y are independent if

$$F(a,b)=F_X(a)F_Y(b) \quad \text{for all } a,b$$

When X and Y are discrete random variables, the condition of independence (2.1) is equivalent to

$$p(x,y)=p_X(x)p_Y(y) \quad \text{for all } x,y \tag{2.2}$$

The equivalence follows because, if **Equation (2.1)** is satisfied, then we obtain **Equation (2.2)** by letting A and B be, respectively, the one-point sets A={x} and B={y}. Furthermore, if **Equation (2.2)** is valid, then for any sets A,B,

$$P\{X\in A, Y\in B\} = \sum_{y\in B}\sum_{x\in A} p(x,y) = \sum_{y\in B}\sum_{x\in A} p_X(x)\, p_Y(y) = \sum_{y\in B} p_Y(y) \sum_{x\in A} p_X(x) = P\{Y\in B\}P\{X\in A\}$$

and **Equation (2.1)** is established.

In the jointly continuous case, the condition of independence is equivalent to

$$f(x,y)=f_X(x)f_Y(y) \quad \text{for all } x,y$$

Thus, loosely speaking, X and Y are independent if knowing the value of one does not change the distribution of the other. Random variables that are not independent are said to be *dependent*.

Example 2a

Suppose that n+m independent trials having a common probability of success p are performed. If X is the number of successes in the first n trials, and Y is the number of successes in the final m trials, then X and Y are independent, since knowing the number of successes in the first n trials does not affect the distribution of the number of successes in the final m trials (by the assumption of independent trials). In fact, for integral x and y,

$$P\{X=x,Y=y\} = \binom{n}{x} p^x (1-p)^{n-x} \binom{m}{y} p^y (1-p)^{m-y} \qquad 0\le x\le n,\ 0\le y\le m = P\{X=x\}P\{Y=y\}$$

In contrast, X and Z will be dependent, where Z is the total number of successes in the n+m trials. (Why?)

Example 2b

Suppose that the number of people who enter a post office on a given day is a Poisson random variable with parameter λ. Show that if each person who enters the post office is a male with probability p and a female with probability 1−p, then the number of males and females entering the post office are independent Poisson random variables with respective parameters λp and λ(1−p).

Solution

Let X and Y denote, respectively, the number of males and females that enter the post office. We shall show the independence of X and Y by establishing **Equation (2.2)**. To obtain an expression for P{X=i,Y=j}, we condition on whether or not X+Y=i+j. This gives:

$$P\{ X=i,Y=j \} =P\{X=i,Y=j|X+Y=i+j\}P\{ X+Y=i+j \} \quad +P\{X=i,Y=j|X+Y\neq i+j\}P\{ X+Y\neq i+j \}$$

[Note that this equation is merely a special case of the formula P(E)=P(E|F)P(F)+P(E|Fc)P(Fc).]

Since P{X=i,Y=j|X+Y≠i+j} is clearly 0, we obtain

$$P\{X=i,Y=j\}=P\{X=i,Y=j|X+Y=i+j\}P\{X+Y=i+j\} \tag{2.3}$$

Now, because X+Y is the total number of people who enter the post office, it follows, by assumption, that

$$P\{X+Y=i+j\}=e-\lambda\lambda i+j(i+j)! \tag{2.4}$$

Furthermore, given that i+j people do enter the post office, since each person entering will be male with probability p, it follows that the probability that exactly i of them will be male (and thus j of them female) is just the binomial probability (i+j i) p i (1−p) j . That is,

$$P\{X=i,Y=j|X+Y=i+j\}=(i+j\ i\ ) p\ i\ (1-p)\ j \tag{2.5}$$

Substituting **Equations (2.4)** and (**2.5**) into **Equation (2.3)** yields

$$P\{ X=i,Y=j \} =(i+j\ i\ ) p\ i\ (1-p)\ j\ e\ -\lambda\ \lambda\ i+j\ (i+j)! = e\ -\lambda\ (\lambda p)\ i\ i!j!\ [\ \lambda(1-p\ )\ ]\ j = e\ -\lambda\ (\lambda p)\ i\ i!\ e\ -\lambda\ (1-p)\ [\ \lambda(1-p\ )\ ]\ j\ j! \tag{2.6}$$

Hence,

$$P\{X=i\}=e-\lambda p(\lambda p)ii!\sum je-\lambda(1-p)[\lambda(1-p)]jj!=e-\lambda p(\lambda p)ii! \tag{2.7}$$

and similarly,

$$P\{Y=j\}=e-\lambda(1-p)[\lambda(1-p)]jj! \tag{2.8}$$

Equations (2.6), **(2.7)**, and **(2.8)** establish the desired result.

Example 2c

A man and a woman decide to meet at a certain location. If each of them independently arrives at a time uniformly distributed between 12 noon and 1 P.M., find the probability that the first to arrive has to wait longer than 10 minutes.

Solution

If we let X and Y denote, respectively, the time past 12 that the man and the woman arrive, then X and Y are independent random variables, each of which is uniformly distributed over (0, 60). The desired

probability, P{X+10<Y}+P{Y+10<X}, which, by symmetry, equals 2P{X+10<Y}, is obtained as follows:

$$2P\{X+10<Y\} = 2\iint_{x+10<y} f(x,y)\,dx\,dy = 2\iint_{x+10<y} f_X(x) f_Y(y)\,dx\,dy = 2\int_{10}^{60}\int_{0}^{y-10}\left(\frac{1}{60}\right)^2 dx\,dy = 2\left(\frac{1}{60}\right)^2\int_{10}^{60}(y-10)\,dy = \frac{25}{36}$$

Our next example presents the oldest problem dealing with geometrical probabilities. It was first considered and solved by Buffon, a French naturalist of the eighteenth century, and is usually referred to as *Buffon's needle problem*.

Example 2d Buffon's Needle Problem

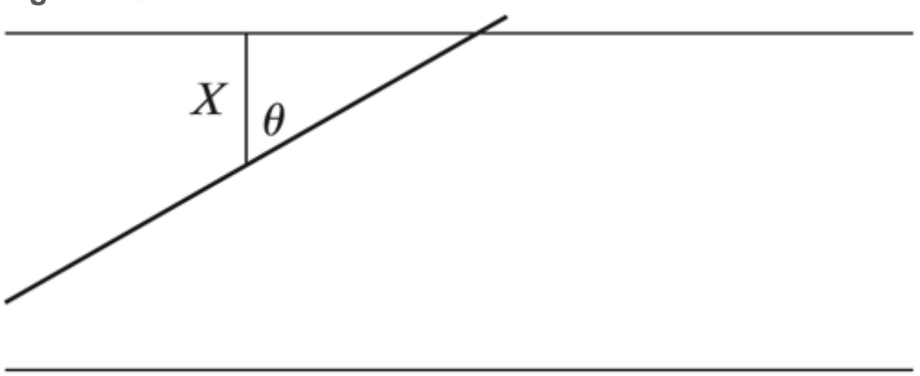
A table is ruled with equidistant parallel lines a distance D apart. A needle of length L, where L≤D, is randomly thrown on the table. What is the probability that the needle will intersect one of the lines (the other possibility being that the needle will be completely contained in the strip between two lines)?

Solution

Let us determine the position of the needle by specifying (1) the distance X from the middle point of the needle to the nearest parallel line and (2) the angle θ between the needle and the projected line of length X. (See **Figure 6.2**.) The needle will intersect a line if the hypotenuse of the right triangle in **Figure 6.2** is less than L/2—that is, if

$$X\cos\theta < \frac{L}{2} \quad \text{or} \quad X < \frac{L}{2}\cos\theta$$

Figure 6.2



As X varies between 0 and D/2 and θ between 0 and π/2, it is reasonable to assume that they are independent, uniformly distributed random variables over these respective ranges. Hence,

$$P\{X < \frac{L}{2}\cos\theta\} = \iint_{x<\frac{L}{2}\cos y} f_X(x) f_\theta(y)\,dx\,dy = \frac{4}{\pi D}\int_0^{\pi/2}\int_0^{\frac{L}{2}\cos y} dx\,dy = \frac{4}{\pi D}\int_0^{\pi/2}\frac{L}{2}\cos y\,dy = \frac{2L}{\pi D}$$

***Example 2e Characterization of the Normal Distribution**

Let X and Y denote the horizontal and vertical miss distances when a bullet is fired at a target, and assume that

1. X and Y are independent continuous random variables having differentiable density functions.
2. The joint density $f(x,y)=f_X(x)f_Y(y)$ of X and Y depends on (x, y) only through x^2+y^2.

Loosely put, assumption 2 states that the probability of the bullet landing on any point of the x–y plane depends only on the distance of the point from the target and not on its angle of orientation. An equivalent way of phrasing this assumption is to say that the joint density function is rotation invariant.

It is a rather interesting fact that assumptions 1 and 2 imply that X and Y are normally distributed random variables. To prove this, note first that the assumptions yield the relation

$$f(x,y)=f_X(x)f_Y(y)=g(x^2+y^2) \tag{2.9}$$

for some function g. Differentiating **Equation (2.9)** with respect to x yields

$$f'_X(x)f_Y(y)=2x\, g'(x^2+y^2) \tag{2.10}$$

Dividing **Equation (2.10)** by **Equation (2.9)** gives

$$f'_X(x)f_X(x)=2x\, g'(x^2+y^2)g(x^2+y^2)$$

or

$$f'_X(x)2x\, f_X(x)=g'(x^2+y^2)g(x^2+y^2) \tag{2.11}$$

Because the value of the left-hand side of **Equation (2.11)** depends only on x, whereas the value of the right-hand side depends on x^2+y^2, it follows that the left-hand side must be the same for all x. To see this, consider any x_1, x_2 and let y_1, y_2 be such that $x_1^2+y_1^2=x_2^2+y_2^2$. Then, from **Equation (2.11)**, we obtain

$$f'_X(x_1)2x_1\, f_X(x_1)=g'(x_1^2+y_1^2)g(x_1^2+y_1^2)=g'(x_2^2+y_2^2)g(x_2^2+y_2^2)=f'_X(x_2)2x_2\, f_X(x_2)$$

Hence,

$$f'_X(x)x\, f_X(x)=c \quad \text{or} \quad \frac{d}{dx}(\log f_X(x))=cx$$

which implies, upon integration of both sides, that

$$\log f_X(x)=a+cx^2/2 \quad \text{or} \quad f_X(x)=ke^{cx^2/2}$$

Since $\int_{-\infty}^{\infty} f_X(x)\, dx=1$, it follows that c is necessarily negative, and we may write $c=-1/\sigma^2$. Thus,

$$f_X(x)=ke^{-x^2/2\sigma^2}$$

That is, X is a normal random variable with parameters $\mu=0$ and σ^2. A similar argument can be applied to $f_Y(y)$ to show that

$$f_Y(y)=\frac{1}{\sqrt{2\pi}\,\sigma}e^{-y^2/2\sigma^2}$$

Furthermore, it follows from assumption 2 that $\sigma_2=\sigma^2$ and that X and Y are thus independent, identically distributed normal random variables with parameters $\mu=0$ and σ^2.

A necessary and sufficient condition for the random variables X and Y to be independent is for their joint probability density function (or joint probability mass function in the discrete case) f(x, y) to factor into two terms, one depending only on x and the other depending only on y.

Proposition 2.1

The continuous (discrete) random variables X and Y are independent if and only if their joint probability density (mass) function can be expressed as

$$f_{X,Y}(x,y)=h(x)\, g(y) \qquad -\infty<x<\infty, -\infty<y<\infty$$

Proof Let us give the proof in the continuous case. First, note that independence implies that the joint density is the product of the marginal densities of X and Y, so the preceding factorization will hold when the random variables are independent. Now, suppose that

$$f_{X,Y}(x,y)=h(x)\,g(y)$$

Then

$$1=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx\,dy=\int_{-\infty}^{\infty} h(x)\,dx\int_{-\infty}^{\infty} g(y)\,dy=C_1C_2$$

where $C_1=\int_{-\infty}^{\infty} h(x)\,dx$ and $C_2=\int_{-\infty}^{\infty} g(y)\,dy$. Also,

$$f_X(x)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy=C_2\,h(x)\qquad f_Y(y)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx=C_1\,g(y)$$

Since $C_1C_2=1$, it follows that

$$f_{X,Y}(x,y)=f_X(x)f_Y(y)$$

and the proof is complete.

Example 2f

If the joint density function of X and Y is

$$f(x,y)=6e^{-2x}e^{-3y}\qquad 0<x<\infty,\ 0<y<\infty$$

and is equal to 0 outside this region, are the random variables independent? What if the joint density function is

$$f(x,y)=24xy\qquad 0<x<1,\ 0<y<1,\ 0<x+y<1$$

and is equal to 0 otherwise?

Solution

In the first instance, the joint density function factors, and thus the random variables, are independent (with one being exponential with rate 2 and the other exponential with rate 3). In the second instance, because the region in which the joint density is nonzero cannot be expressed in the form $x\in A, y\in B$, the joint density does not factor, so the random variables are not independent. This can be seen clearly by letting

$$I(x,y)=\begin{cases}1 & \text{if } 0<x<1,\ 0<y<1,\ 0<x+y<1\\ 0 & \text{otherwise}\end{cases}$$

and writing

$$f(x,y)=24xy\,I(x,y)$$

which clearly does not factor into a part depending only on x and another depending only on y.

The concept of independence may, of course, be defined for more than two random variables. In general, the n random variables $X_1,X_2,\ldots,X_n$ are said to be independent if, for all sets of real numbers $A_1,A_2,\ldots,A_n$,

$$P\{X1\in A1, X2\in A2, \ldots, Xn\in An\} = \prod_{i=1}^{n} P\{Xi\in Ai\}$$

As before, it can be shown that this condition is equivalent to

$$P\{X_1 \le a_1, X_2 \le a_2, \ldots, X_n \le a_n\} = \prod_{i=1}^{n} P\{X_i \le a_i\} \quad \text{for all } a_1, a_2, \ldots, a_n$$

Finally, we say that an infinite collection of random variables is independent if every finite subcollection of them is independent.

Example 2g How can a computer choose a random subset?

Most computers are able to generate the value of, or *simulate*, a uniform (0, 1) random variable by means of a built-in subroutine that (to a high degree of approximation) produces such "random numbers." As a result, it is quite easy for a computer to simulate an indicator (that is, a Bernoulli) random variable. Suppose I is an indicator variable such that

$$P\{I=1\} = p = 1 - P\{I=0\}$$

The computer can simulate I by choosing a uniform (0, 1) random number U and then letting

$$I = \begin{cases} 1 & \text{if } U < p \\ 0 & \text{if } U \ge p \end{cases}$$

Suppose that we are interested in having the computer select k, $k \le n$, of the numbers $1, 2, \ldots, n$ in such a way that each of the $\binom{n}{k}$ subsets of size k is equally likely to be chosen. We now present a method that will enable the computer to solve this task. To generate such a subset, we will first simulate, in sequence, n indicator variables $I_1, I_2, \ldots, I_n$, of which exactly k will equal 1. Those i for which $I_i = 1$ will then constitute the desired subset.

To generate the random variables $I_1, \ldots, I_n$, start by simulating n independent uniform (0, 1) random variables $U_1, U_2, \ldots, U_n$. Now define

$$I_1 = \begin{cases} 1 & \text{if } U_1 < \frac{k}{n} \\ 0 & \text{otherwise} \end{cases}$$

and then, once $I_1, \ldots, I_i$ are determined, recursively set

$$I_{i+1} = \begin{cases} 1 & \text{if } U_{i+1} < \frac{k - (I_1 + \cdots + I_i)}{n-i} \\ 0 & \text{otherwise} \end{cases}$$

In words, at the (i+1)th stage, we set I_{i+1} equal to 1 (and thus put i+1 into the desired subset) with a probability equal to the remaining number of places in the subset (namely, $k - \sum_{j=1}^{i} I_j$), divided by the remaining number of possibilities (namely, n−i). Hence, the joint distribution of $I_1, I_2, \ldots, I_n$ is determined from

$$P\{I_1 = 1\} = \frac{k}{n} \quad P\{I_{i+1} = 1 \mid I_1, \ldots, I_i\} = \frac{k - \sum_{j=1}^{i} I_j}{n-i} \quad 1 < i < n$$

The proof that the preceding formula results in all subsets of size k being equally likely to be chosen is by induction on k+n. It is immediate when k+n=2 (that is, when k=1, n=1), so assume it to be true whenever $k+n \le l$. Now, suppose that k+n=l+1, and consider any subset of size k—say, $i_1 \le i_2 \le \cdots \le i_k$—and consider the following two cases.

Case 1: $i_1 = 1$

$$P\{I_1 = I_{i_2} = \cdots = I_{i_k} = 1, I_j = 0 \text{ otherwise}\} = P\{I_1 = 1\} P\{I_{i_2} = \cdots = I_{i_k} = 1, I_j = 0 \text{ otherwise} \mid I_1 = 1\}$$

Now given that $I_1 = 1$, the remaining elements of the subset are chosen as if a subset of size k−1 were to

be chosen from the n−1 elements 2,3,… ,n. Hence, by the induction hypothesis, the conditional probability that this will result in a given subset of size k−1 being selected is 1/(n−1k−1). Hence,

P{ I 1 = I i 2 =⋯= I i k =1, I j =0 otherwise} = k n 1 (n−1 k−1) = 1 (n k)

Case 2: i 1 ≠1

P{ I i1 = I i 2 =⋯= I i k =1, I j =0 otherwise} =P{ I i 1 =⋯= I i k =1, I j =0 otherwise| I 1 =0}P{ I 1 =0 } = 1 (n−1 k) (1− k n)= 1 (n k)

where the induction hypothesis was used to evaluate the preceding conditional probability.

Thus, in all cases, the probability that a given subset of size k will be the subset chosen is 1/(nk).

Remark The foregoing method for generating a random subset has a very low memory requirement. A faster algorithm that requires somewhat more memory is presented in **Section 10.1**. (The latter algorithm uses the last k elements of a random permutation of 1,2,… ,n.)

Example 2h

Let X, Y, Z be independent and uniformly distributed over (0, 1). Compute P{X≥YZ}.

Solution

Since

f X,Y,Z (x,y,z) = f X (x) f Y (y) f Z (z) =1, 0≤x≤1, 0≤y≤1, 0≤z≤1

we have

P{ X≥YZ } = ∫∫∫ x≥yz f X, Y, Z (x,y,z) dx dy dz = ∫ 0 1 ∫ 0 1 ∫ yz 1 dx dy dz = ∫ 0 1 ∫ 0 1 (1−yz) dy dz = ∫ 0 1 (1− z 2) dz = 3 4

Example 2i Probabilistic Interpretation of Half-Life

Let N(t) denote the number of nuclei contained in a radioactive mass of material at time t. The concept of half-life is often defined in a deterministic fashion by stating this it is an empirical fact that, for some value h, called the *half-life*,

N(t)=2−t/hN(0) t>0

[Note that N(h)=N(0)/2.] Since the preceding implies that, for any nonnegative s and t,

N(t+s)=2−(s+t)/hN(0)=2−t/hN(s)

it follows that no matter how much time s has already elapsed, in an additional time t, the number of existing nuclei will decrease by the factor 2−t/h.

Because the deterministic relationship just given results from observations of radioactive masses containing huge numbers of nuclei, it would seem that it might be consistent with a probabilistic interpretation. The clue to deriving the appropriate probability model for half-life resides in the empirical observation that the proportion of decay in any time interval depends neither on the total number of nuclei at the beginning of the interval nor on the location of this interval [since N(t+s)/N(s) depends neither on N(s) nor on s]. Thus, it appears that the individual nuclei act independently and with a memoryless life distribution. Consequently, since the unique life distribution that is memoryless is the

exponential distribution, and since exactly one-half of a given amount of mass decays every h time units, we propose the following probabilistic model for radioactive decay.

Probabilistic interpretation of the half-life *h*: The lifetimes of the individual nuclei are independent random variables having a life distribution that is exponential with median equal to h. That is, if L represents the lifetime of a given nucleus, then

$$P\{L<t\}=1-2^{-t/h}$$

(Because $P\{L<h\}=12$ and the preceding can be written as

$$P\{L<t\}=1-\exp\{-t\log 2h\}$$

it can be seen that L indeed has an exponential distribution with median h.)

Note that under the probabilistic interpretation of half-life just given, if one starts with N(0) nuclei at time 0, then N(t), the number of nuclei that remain at time t will have a binomial distribution with parameters n=N(0) and $p=2^{-t/h}$. Results of **Chapter 8** will show that this interpretation of half-life is consistent with the deterministic model when considering the proportion of a large number of nuclei that decay over a given time frame. However, the difference between the deterministic and probabilistic interpretation becomes apparent when one considers the actual number of decayed nuclei. We will now indicate this with regard to the question of whether protons decay.

There is some controversy over whether or not protons decay. Indeed, one theory predicts that protons should decay with a half-life of about h=1030 years. To check this prediction empirically, it has been suggested that one follow a large number of protons for, say, one or two years and determine whether any of them decay within that period. (Clearly, it would not be feasible to follow a mass of protons for 1030 years to see whether one-half of it decays.) Let us suppose that we are able to keep track of N(0)=1030 protons for c years. The number of decays predicted by the deterministic model would then be given by

$$N(0)-N(c) = h(1-2^{-c/h}) = 1-2^{-c/h}\ 1/h \approx \lim_{x\to 0} \frac{1-2^{-cx}}{x} \quad \text{since } 1\ h = 10^{-30} \approx 0 = \lim_{x\to 0} \frac{(c\,2^{-cx}\log 2)}{1} \text{ by L'Hôpital's rule } = c\log 2 \approx .6931c$$

For instance, the deterministic model predicts that in 2 years there should be 1.3863 decays, and it would thus appear to be a serious blow to the hypothesis that protons decay with a half-life of 1030 years if no decays are observed over those 2 years.

Let us now contrast the conclusions just drawn with those obtained from the probabilistic model. Again, let us consider the hypothesis that the half-life of protons is h=1030 years, and suppose that we follow h protons for c years. Since there is a huge number of independent protons, each of which will have a very small probability of decaying within this time period, it follows that the number of protons that decay will have (to a very strong approximation) a Poisson distribution with parameter equal to $h(1-2^{-c/h}) \approx c\log 2$. Thus,

$$P\{0 \text{ decays}\} = e^{-c\log 2} = e^{-\log(2^c)} = 1/2^c$$

and, in general,

$$P\{n \text{ decays}\} = 2^{-c}[c\log 2]^n/n! \qquad n \geq 0$$

Thus, we see that even though the average number of decays over 2 years is (as predicted by the deterministic model) 1.3863, there is 1 chance in 4 that there will not be any decays, thereby indicating that such a result in no way invalidates the original hypothesis of proton decay.

Remarks *Independence is a symmetric relation*. The random variables X and Y are independent if their joint density function (or mass function in the discrete case) is the product of their individual density (or mass) functions. Therefore, to say that X is independent of Y is equivalent to saying that Y is independent of X — or just that X and Y are independent. As a result, in considering whether X is independent of Y in situations where it is not at all intuitive that knowing the value of Y will not change the probabilities concerning X, it can be beneficial to interchange the roles of X and Y and ask instead whether Y is independent of X. The next example illustrates this point.

Example 2j

If the initial throw of the dice in the game of craps results in the sum of the dice equaling 4, then the player will continue to throw the dice until the sum is either 4 or 7. If this sum is 4, then the player wins, and if it is 7, then the player loses. Let N denote the number of throws needed until either 4 or 7 appears, and let X denote the value (either 4 or 7) of the final throw. Is N independent of X? That is, does knowing which of 4 or 7 occurs first affect the distribution of the number of throws needed until that number appears? Most people do not find the answer to this question to be intuitively obvious. However, suppose that we turn it around and ask whether X is independent of N. That is, does knowing how many throws it takes to obtain a sum of either 4 or 7 affect the probability that that sum is equal to 4? For instance, suppose we know that it takes n throws of the dice to obtain a sum of either 4 or 7. Does this affect the probability distribution of the final sum? Clearly not, since all that is important is that its value is either 4 or 7, and the fact that none of the first n−1 throws were either 4 or 7 does not change the probabilities for the nth throw. Thus, we can conclude that X is independent of N, or equivalently, that N is independent of X.

As another example, let X1,X2,… be a sequence of independent and identically distributed continuous random variables, and suppose that we observe these random variables in sequence. If Xn>Xi for each i=1,… ,n−1, then we say that Xn is a *record value*. That is, each random variable that is larger than all those preceding it is called a record value. Let An denote the event that Xn is a record value. Is An+1 independent of An? That is, does knowing that the nth random variable is the largest of the first n change the probability that the (n+1) random variable is the largest of the first n+1? While it is true that An+1 is independent of An, this may not be intuitively obvious. However, if we turn the question around and ask whether An is independent of An+1, then the result is more easily understood. For knowing that the (n+1) value is larger than X1,… ,Xn clearly gives us no information about the relative size of Xn among the first n random variables. Indeed, by symmetry, it is clear that each of these n random variables is equally likely to be the largest of this set, so P(An|An+1)=P(An)=1/n. Hence, we can conclude that An and An+1 are independent events.

Remark It follows from the identity

$$P\{X_1 \le a_1, \ldots, X_n \le a_n\} = P\{X_1 \le a_1\} P\{X_2 \le a_2 \mid X_1 \le a_1\} \cdots P\{X_n \le a_n \mid X_1 \le a_1, \ldots, X_{n-1} \le a_{n-1}\}$$

that the independence of X1,… ,Xn can be established sequentially. That is, we can show that these random variables are independent by showing that

$$X_2 \text{ is independent of } X_1 \quad X_3 \text{ is independent of } X_1, X_2 \quad X_4 \text{ is independent of } X_1, X_2, X_3 \vdots X_n \text{ is independent of } X_1, \ldots, X_{n-1}$$

6.3 Sums of Independent Random Variables

It is often important to be able to calculate the distribution of X+Y from the distributions of X and Y when X and Y are independent. Suppose that X and Y are independent, continuous random variables having probability density functions fX and fY. The cumulative distribution function of X+Y is obtained as follows:

$$F_{X+Y}(a) = P\{X+Y \le a\} = \iint_{x+y \le a} f_X(x) f_Y(y)\, dx\, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{a-y} f_X(x) f_Y(y)\, dx\, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{a-y} f_X(x) dx\, f_Y(y)\, dy = \int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y)\, dy \quad (3.1)$$

The cumulative distribution function FX+Y is called the *convolution* of the distributions FX and FY (the cumulative distribution functions of X and Y, respectively).

By differentiating **Equation (3.1)**, we find that the probability density function fX+Y of X+Y is given by

$$f_{X+Y}(a) = \frac{d}{da} \int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y) dy = \int_{-\infty}^{\infty} \frac{d}{da} F_X(a-y)\, f_Y(y)\, dy = \int_{-\infty}^{\infty} f_X(a-y)\, f_Y(y)\, dy \quad (3.2)$$

6.3.1 Identically Distributed Uniform Random Variables

It is not difficult to determine the density function of the sum of two independent uniform (0,1) random variables.

Example 3a Sum of two independent uniform random variables

If X and Y are independent random variables, both uniformly distributed on (0, 1), calculate the probability density of X+Y.

Solution

From **Equation (3.2)**, since

$$f_X(a) = f_Y(a) = \begin{cases} 1 & 0 < a < 1 \\ 0 & \text{otherwise} \end{cases}$$

we obtain

$$f_{X+Y}(a) = \int_0^1 f_X(a-y)\, dy$$

For $0 \le a \le 1$, this yields

$$f_{X+Y}(a) = \int_0^a dy = a$$

For $1 < a < 2$, we get

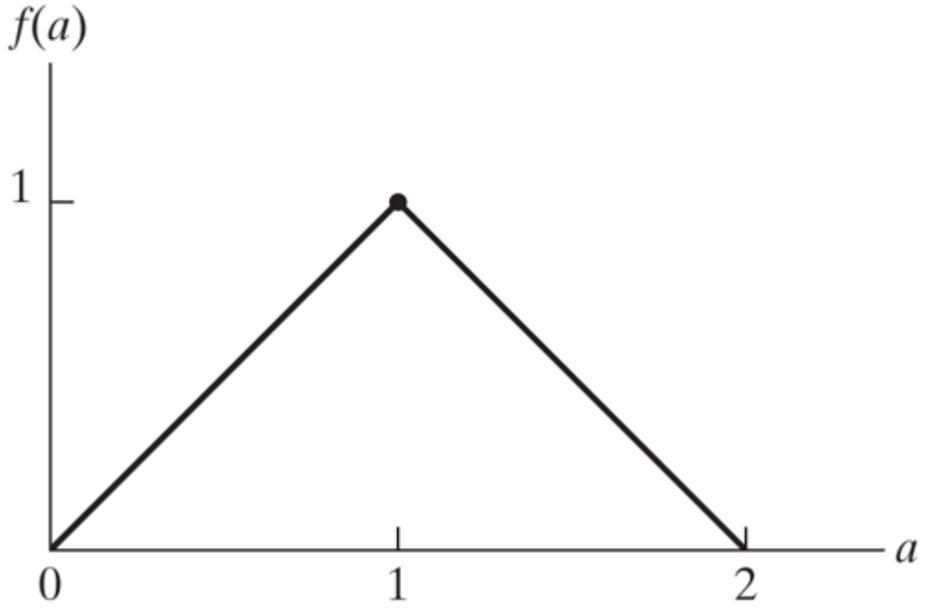
$$f_{X+Y}(a) = \int_{a-1}^1 dy = 2-a$$

Hence,

$$f_{X+Y}(a) = \begin{cases} a & 0 \le a \le 1 \\ 2-a & 1 < a < 2 \\ 0 & \text{otherwise} \end{cases}$$

Because of the shape of its density function (see **Figure 6.3**), the random variable X+Y is said to have a *triangular* distribution.

Figure 6.3 Triangular density function.



Now, suppose that X1,X2,… ,Xn are independent uniform (0,1) random variables, and let

Fn(x)=P{X1+… +Xn≤x}

Whereas a general formula for Fn(x) is messy, it has a particularly nice form when x≤1. Indeed, we now use mathematical induction to prove that

Fn(x)=xn/n! , 0≤x≤1

Because the proceeding equation is true for n=1, assume that

Fn−1(x)=xn−1/(n−1)! , 0≤x≤1

Now, writing

∑i=1nXi=∑i=1n−1Xi+Xn

and using the fact that the Xi are all nonnegative, we see from **Equation (3.1)** that, for 0≤x≤1,

F n (x) = ∫ 0 1 F n−1 (x−y) f X n (y)dy = 1 (n−1)! ∫ 0 x (x−y) n−1 dy by the induction hypothesis = 1 (n−1)! ∫ 0 x w n−1 dw (by w=x−y) = x n /n!

which completes the proof.

For an interesting application of the preceding formula, let us use it to determine the expected number of independent uniform (0,1) random variables that need to be summed to exceed 1. That is, with X1,X2,… being independent uniform (0,1) random variables, we want to determine E[N], where

N=min{n:X1+… +Xn>1}

Noting that N is greater than n>0 if and only if X 1 +…+ X n ≤1, we see that

P{N>n}=Fn(1)=1/n! , n>0

Because

P{N>0}=1=1/0!

we see that, for n>0,

P{ N=n }=P{N>n−1}−P{N>n}= 1 (n−1)! − 1 n! = n−1 n!

Therefore,

E[N] = ∑ n=1 ∞ n(n−1) n! = ∑ n=2 ∞ 1 (n−2)! =e

That is, the mean number of independent uniform (0,1) random variables that must be summed for the sum to exceed 1 is equal to e.

6.3.2 Gamma Random Variables

Recall that a gamma random variable has a density of the form

f(y)=λe−λy(λy)t−1Γ(t) 0<y<∞

An important property of this family of distributions is that for a fixed value of λ, it is closed under convolutions.

> ***Proposition 3.1***
>
> If X and Y are independent gamma random variables with respective parameters (s,λ) and (t,λ), then X+Y is a gamma random variable with parameters (s+t,λ).
>
> **Proof** Using **Equation (3.2)**, we obtain
>
> f X+Y (a) = 1 Γ(s)Γ(t) ∫ 0 a λ e −λ(a−y) [λ(a−y)] s−1 λ e −λy (λy) t−1 dy =K e −λa ∫ 0 a (a−y) s−1 y t−1 dy =K e −λa a s+t−1 ∫ 0 1 (1−x) s−1 x t−1 dx by letting x= y a =C e −λa a s+t−1
>
> where C is a constant that does not depend on a. But, as the preceding is a density function and thus must integrate to 1, the value of C is determined, and we have
>
> fX+Y(a)=λe−λa(λa)s+t−1Γ(s+t)
>
> Hence, the result is proved.

It is now a simple matter to establish, by using **Proposition 3.1** and induction, that if Xi,i=1,… ,n are independent gamma random variables with respective parameters (ti,λ),i=1,… ,n, then ∑i=1nXi is gamma with parameters (∑i=1nti,λ). We leave the proof of this statement as an exercise.

> ***Example 3b***
>
> Let X1,X2,… ,Xn be n independent exponential random variables, each having parameter λ. Then, since an exponential random variable with parameter λ is the same as a gamma random variable with parameters (1,λ), it follows from **Proposition 3.1** that X1+X2+⋯+Xn is a gamma random variable with parameters (n,λ).

If Z1,Z2,… ,Zn are independent standard normal random variables, then ∑ i=1 n Z i 2 is said to have the *chi-squared* (sometimes seen as χ2) distribution with n degrees of freedom. Let us compute the density function of Y. When n=1,Y=Z21, and from **Example 7b** of **Chapter 5**, we see that its probability density function is given by

$$f_{Z^2}(y) = \frac{1}{2\sqrt{y}}\left[f_Z(\sqrt{y}) + f_Z(-\sqrt{y})\right] = \frac{1}{2\sqrt{y}}\frac{2}{\sqrt{2\pi}}e^{-y/2} = \frac{1}{2}e^{-y/2}(y/2)^{1/2-1}\pi$$

But we recognize the preceding as the gamma distribution with parameters (12,12). [A by-product of this analysis is that $\Gamma(12)=\pi$.] But since each Z_i^2 is gamma (12,12), it follows from **Proposition 3.1** that the chi-squared distribution with n degrees of freedom is just the gamma distribution with parameters (n/2,12) and hence has a probability density function given by

$$f_Y(y) = \frac{1}{2}e^{-y/2}\left(\frac{y}{2}\right)^{n/2-1}\frac{1}{\Gamma\left(\frac{n}{2}\right)} \quad y>0 = \frac{e^{-y/2}y^{n/2-1}}{2^{n/2}\Gamma\left(\frac{n}{2}\right)} \quad y>0$$

When n is an even integer, $\Gamma(n/2)=[(n/2)-1]!$, whereas when n is odd, $\Gamma(n/2)$ can be obtained from iterating the relationship $\Gamma(t)=(t-1)\Gamma(t-1)$ and then using the previously obtained result that $\Gamma(12)=\sqrt{\pi}$. [For instance, $\Gamma(52)=32\Gamma(32)=3212\Gamma(12)=34\sqrt{\pi}$.]

In practice, the chi-squared distribution often arises as the distribution of the square of the error involved when one attempts to hit a target in n-dimensional space when the coordinate errors are taken to be independent standard normal random variables. It is also important in statistical analysis.

6.3.3 Normal Random Variables

We can also use **Equation (3.2)** to prove the following important result about normal random variables.

Proposition 3.2

If $X_i, i=1,\ldots,n$, are independent random variables that are normally distributed with respective parameters $\mu_i,\sigma_i^2, i=1,\ldots,n$, then $\sum_{i=1}^{n}X_i$ is normally distributed with parameters $\sum_{i=1}^{n}\mu_i$ and $\sum_{i=1}^{n}\sigma_i^2$.

Proof of Proposition 3.2: To begin, let X and Y be independent normal random variables with X having mean 0 and variance σ^2 and Y having mean 0 and variance 1. We will determine the density function of X+Y by utilizing **Equation (3.2)**. Now, with

$$c=\frac{1}{2\sigma^2}+\frac{1}{2}=\frac{1+\sigma^2}{2\sigma^2}$$

we have

$$f_X(a-y)f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\left\{-\frac{(a-y)^2}{2\sigma^2}\right\}\frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{y^2}{2}\right\} = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2\sigma^2}\right\}\exp\left\{-cy^2+\frac{ay}{\sigma^2}\right\} = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2\sigma^2}-c\left(y^2-\frac{2ay}{1+\sigma^2}\right)\right\}$$

where the preceding follows because $\frac{2c}{1+\sigma^2}=\frac{1}{\sigma^2}$. Now,

$$\frac{a^2}{2\sigma^2}+c\left(y^2-\frac{2ya}{1+\sigma^2}\right) = \frac{a^2}{2\sigma^2}+c\left(y-\frac{a}{1+\sigma^2}\right)^2-c\frac{a^2}{(1+\sigma^2)^2} = \frac{a^2}{2\sigma^2}+c\left(y-\frac{a}{1+\sigma^2}\right)^2-\frac{a^2}{2\sigma^2(1+\sigma^2)} = \frac{a^2}{2\sigma^2}\left(1-\frac{1}{1+\sigma^2}\right)+c\left(y-\frac{a}{1+\sigma^2}\right)^2 = \frac{a^2}{2(1+\sigma^2)}+c\left(y-\frac{a}{1+\sigma^2}\right)^2$$

Hence,

$$f_X(a-y)f_Y(y) = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}\exp\left\{-c\left(y-\frac{a}{1+\sigma^2}\right)^2\right\}$$

From **Equation (3.2)**, we obtain that

$$f_{X+Y}(a) = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}\int_{-\infty}^{\infty}\exp\left\{-c\left(y-\frac{a}{1+\sigma^2}\right)^2\right\}dy = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}\int_{-\infty}^{\infty}\exp\{-cx^2\}\,dx = C\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}$$

where C does not depend on a. But this implies that X+Y is normal with mean 0 and variance 1+σ2.

Now, suppose that X1 and X2 are independent normal random variables with Xi having mean μi and variance σ i 2 , i=1,2. Then

X1+X2=σ2(X1−μ1σ2+X2−μ2σ2)+μ1+μ2

But since (X1−μ1)/σ2 is normal with mean 0 and variance σ12/σ22, and (X2−μ2)/σ2 is normal with mean 0 and variance 1, it follows from our previous result that (X1−μ1)/σ2+(X2−μ2)/σ2 is normal with mean 0 and variance 1+ σ 1 2 / σ 2 2 , implying that X1+X2 is normal with mean μ1+μ2 and variance σ 2 2 (1+ σ 1 2 / σ 2 2)= σ 1 2 + σ 2 2 .

Thus, **Proposition 3.2** is established when n=2. The general case now follows by induction. That is, assume that **Proposition 3.2** is true when there are n−1 random variables. Now consider the case of n, and write

∑i=1n Xi=∑i=1n−1Xi+Xn

By the induction hypothesis, ∑i=1n−1Xi is normal with mean ∑i=1n−1μi and variance ∑ i−1 n−1 σ i 2 . Therefore, by the result for n=2, ∑i=1nXi is normal with mean ∑i=1nμi and variance ∑ i−1 n σ i 2 .

Example 3c

A basketball team will play a 44-game season. Twenty-six of these games are against class A teams and 18 are against class B teams. Suppose that the team will win each game against a class A team with probability .4 and will win each game against a class B team with probability .7. Suppose also that the results of the different games are independent. Approximate the probability that

a. the team wins 25 games or more;
b. the team wins more games against class A teams than it does against class B teams.

Solution

a. Let XA and XB respectively denote the number of games the team wins against class A and against class B teams. Note that XA and XB are independent binomial random variables and E[X A]=26(.4)=10.4 Var(X A)=26(.4)(.6)=6.24 E[X B]=18(.7)=12.6 Var(X B)=18(.7)(.3)=3.78

   By the normal approximation to the binomial, XA and XB will have approximately the same distribution as would independent normal random variables with the preceding expected values and variances. Hence, by **Proposition 3.2**, XA+XB will have approximately a normal distribution with mean 23 and variance 10.02. Therefore, letting Z denote a standard normal random variable, we have
   P{ X A + X B ≥25 } =P{ X A + X B ≥24.5 } =P{ X A + X B −23 10.02 ≥ 24.5−23 10.02 } ≈P{ Z≥ 1.5 10.02 } ≈1−P{Z<.4739} ≈.3178

b. We note that XA−XB will have approximately a normal distribution with mean −2.2 and variance 10.02. Hence,
   P{ X A − X B ≥1 } =P{ X A − X B ≥.5 } =P{ X A − X B +2.2 10.02 ≥ .5+2.2 10.02 } ≈P{ Z≥ 2.7 10.02 } ≈1−P{Z<.8530} ≈.1968

Therefore, there is approximately a 31.78 percent chance that the team will win at least 25 games and approximately a 19.68 percent chance that it will win more games against class A teams than against class B teams.

The random variable Y is said to be a *lognormal* random variable with parameters μ and σ if log (Y) is a normal random variable with mean μ and variance σ2. That is, Y is lognormal if it can be expressed as

Y=eX

where X is a normal random variable.

Example 3d

Starting at some fixed time, let S(n) denote the price of a certain security at the end of n additional weeks, n≥1. A popular model for the evolution of these prices assumes that the price ratios S(n)/S(n−1),n≥1, are independent and identically distributed lognormal random variables. Assuming this model, with parameters μ=.0165,σ=.0730, what is the probability that

a. the price of the security increases over each of the next two weeks?
b. the price at the end of two weeks is higher than it is today?

Solution

Let Z be a standard normal random variable. To solve part (a), we use the fact that log(x) increases in x to conclude that x>1 if and only if log (x)> log (1)=0. As a result, we have

P{ S(1) S(0) >1} =P{log(S(1) S(0))>0} =P{Z> −.0165 .0730 } =P{Z<.2260} =.5894

In other words, the probability that the price is up after one week is .5894. Since the successive price ratios are independent, the probability that the price increases over each of the next two weeks is (.5894)2=.3474.

To solve part (b), we reason as follows:

P{ S(2) S(0) >1} =P{ S(2) S(1) S(1) S(0) >1} =P{log(S(2) S(1))+log(S(1) S(0))>0}

However, log (S(2)S(1))+log (S(1)S(0)), being the sum of two independent normal random variables with a common mean .0165 and a common standard deviation .0730, is a normal random variable with mean .0330 and variance 2(.0730)2. Consequently,

P{ S(2) S(0) >1} =P{Z> −.0330 .0730 2 } =P{Z<.31965} =.6254

6.3.4 Poisson and Binomial Random Variables

Rather than attempt to derive a general expression for the distribution of X+Y in the discrete case, we shall consider some examples.

Example 3e Sums of independent poisson random variables

If X and Y are independent Poisson random variables with respective parameters λ1 and λ2, compute the distribution of X+Y.

Solution

Because the event {X+Y=n} may be written as the union of the disjoint events {X=k,Y=n−k},0≤k≤n, we have

P{ X+Y=n } = ∑ k=0 n P{ X=k,Y=n−k } = ∑ k=0 n P{ X=k }P{ Y=n−k } = ∑ k=0 n e − λ 1 λ k 1 k! e − λ 2 λ n−k 2 (n−k)! = e −(λ 1 + λ 2) ∑ k=0 n λ k 1 λ n−k 2 k!(n−k)! = e −(λ 1 + λ 2) n! ∑ k=0 n n! k!(n−k)! λ k 1

λ n−k 2 = e −(λ 1 + λ 2) n! (λ 1 + λ 2) n

Thus, X+X has a Poisson distribution with parameter λ1+λ2.

Example 3f Sums of independent binomial random variables

Let X and Y be independent binomial random variables with respective parameters (n, p) and (m, p). Calculate the distribution of X+Y.

Solution

Recalling the interpretation of a binomial random variable, and without any computation at all, we can immediately conclude that X+Y is binomial with parameters (n+m,p). This follows because X represents the number of successes in n independent trials, each of which results in a success with probability p; similarly, Y represents the number of successes in m independent trials, each of which results in a success with probability p. Hence, given that X and Y are assumed independent, it follows that X+Y represents the number of successes in n+m independent trials when each trial has a probability p of resulting in a success. Therefore, X+Y is a binomial random variable with parameters (n+m,p). To check this conclusion analytically, note that

P{ X+Y=k } = ∑ i=0 n P{ X=i,Y=k−i } = ∑ i=0 n P{ X=i }P{ Y=k−i } = ∑ i=0 n (n i) p i q n−i (m k−i) p k−i q m−k+i

where q=1−p and where (rj)=0 when j<0. Thus,

P{X+Y=k}=pkqn+m−k∑i=0n(ni)(mk−i)

and the conclusion follows upon application of the combinatorial identity

(n+mk)=∑i=0n(ni)(mk−i)

6.4 Conditional Distributions: Discrete Case

Recall that for any two events E and F, the conditional probability of E given F is defined, provided that P(F)>0, by

P(E|F)=P(EF)P(F)

Hence, if X and Y are discrete random variables, it is natural to define the conditional probability mass function of X given that Y=y, by

p X|Y (x|y) =P{X=x|Y=y} = P{ X=x,Y=y } P{ Y=y } = p(x,y) p Y (y)

for all values of y such that pY(y)>0. Similarly, the conditional probability distribution function of X given that Y=y is defined, for all y such that pY(y)>0, by

F X|Y (x|y) =P{X≤x|Y=y} = ∑ a≤x p X|Y (a|y)

In other words, the definitions are exactly the same as in the unconditional case, except that everything is now conditional on the event that Y=y. If X is independent of Y, then the conditional mass function and the distribution function are the same as the respective unconditional ones. This follows because if X is independent of Y, then

p X|Y (x|y) =P{X=x|Y=y} = P{ X=x,Y=y } P{ Y=y } = P{ X=x }P{ Y=y } P{ Y=y } =P{ X=x }

Example 4a

Suppose that p(x,y), the joint probability mass function of X and Y, is given by

p(0,0)=.4 p(0,1)=.2 p(1,0)=.1 p(1,1)=.3

Calculate the conditional probability mass function of X given that Y=1.

Solution

We first note that

pY(1)=∑xp(x,1)=p(0,1)+p(1,1)=.5

Hence,

pX|Y(0|1)=p(0,1)pY(1)=25

and

pX|Y(1|1)=p(1,1)pY(1)=35

Example 4b

If X and Y are independent Poisson random variables with respective parameters λ1 and λ2, calculate the conditional distribution of X given that X+Y=n.

Solution

We calculate the conditional probability mass function of X given that X+Y=n as follows:

$$P\{X=k|X+Y=n\} = P\{ X=k,X+Y=n \} P\{ X+Y=n \} = P\{ X=k,Y=n-k \} P\{ X+Y=n \} = P\{ X=k \}P\{ Y=n-k \} P\{ X+Y=n \}$$

where the last equality follows from the assumed independence of X and Y. Recalling (**Example 3e**) that X+Y has a Poisson distribution with parameter λ1+λ2, we see that the preceding equals

$$P\{X=k|X+Y=n = e - \lambda 1 \lambda 1 k k! e - \lambda 2 \lambda 2 n-k (n-k)! [e -(\lambda 1 + \lambda 2) (\lambda 1 + \lambda 2) n n!] -1 = n! (n-k)! \lambda 1 k \lambda 2 n-k (\lambda 1 + \lambda 2) n =(n k) (\lambda 1 \lambda 1 + \lambda 2) k (\lambda 2 \lambda 1 + \lambda 2) n-k$$

In other words, the conditional distribution of X given that X+Y=n is the binomial distribution with parameters n and λ1/(λ1+λ2).

We can also talk about joint conditional distributions, as is indicated in the next two examples.

Example 4c

Consider the multinomial distribution with joint probability mass function

$$P\{ X i = n i ,i=1,\dots ,k\}= n! n 1 !\cdots n k ! p 1 n 1 \cdots p k n k , \quad n i \geq 0, \ \sum i=1 k n i =n$$

Such a mass function results when n independent trials are performed, with each trial resulting in outcome i with probability pi, ∑ i=1 k p i =1. The random variables Xi, i=1,… ,k, represent, respectively, the number of trials that result in outcome i, i=1,… ,k. Suppose we are given that nj of the trials resulted in outcome j, for j=r+1,… ,k, where Σ j=r+1 k n j =m≤n. Then, because each of the other n−m trials must have resulted in one of the outcomes 1,… ,r, it would seem that the conditional distribution of X1,… ,Xr is the multinomial distribution on n−m trials with respective trial outcome probabilities

$$P\{\text{outcome } i|\text{outcome is not any of } r+1,\dots ,k\}=piFr, i=1,\dots ,r$$

where F r = Σ i=1 k p i is the probability that a trial results in one of the outcomes 1,… ,r.

Solution

To verify this intuition, let n1,… ,nr, be such that ∑i=1rni=n−m. Then

$$P\{ X 1 = n 1 ,\dots , X r = n r | X r+1 = n r+1 ,\dots X k = n k \} = P\{ X 1 = n 1 , \dots , X k = n k \} P\{ X r+1 = n r+1 , \dots , X k = n k \} = n! n!\cdots n k ! p 1 n 1 \cdots p r n r p r+1 n r+1 \cdots p k n k n! (n-m)! n r+1 !\cdots n k ! F r n-m p r+1 n r+1 \cdots p k n k$$

where the probability in the denominator was obtained by regarding outcomes 1,… ,r as a single outcome having probability Fr, thus showing that the probability is a multinomial probability on n trials with outcome probabilities Fr,pr+1,… ,pk. Because ∑i=1rni=n−m, the preceding can be written as

$$P\{ X 1 = n 1 ,\dots , X r = n r | X r+1 = n r+1 ,\dots X k = n k \} = (n-m)! n 1 !\cdots n r ! (p 1 F r) n 1 \cdots (p r F r) n r$$

and our intuition is upheld.

Example 4d

Consider n independent trials, with each trial being a success with probability p. Given a total of k successes, show that all possible orderings of the k successes and n−k failures are equally likely.

Solution

We want to show that given a total of k successes, each of the (n k) possible orderings of k successes and n−k failures is equally likely. Let X denote the number of successes, and consider any ordering of k successes and n−k failures, say, o=(s,… ,s,f,… ,f). Then

P(o|X=k) = P(o,X=k) P(X=k) = P(o) P(X=k) = P k (1−p) n−k (n k) p k (1−p) n−k = 1 (n k)

6.5 Conditional Distributions: Continuous Case

If X and Y have a joint probability density function f(x,y), then the conditional probability density function of X given that Y=y is defined, for all values of y such that fY(y)>0, by

fX|Y(x|y)=f(x,y)fY(y)

To motivate this definition, multiply the left-hand side by dx and the right-hand side by (dx dy)/dy to obtain

f X|Y (x|y) dx = f(x,y) dx dy f Y (y) dy ≈ P{ x≤X≤x+dx,y≤Y≤y+dy } P{ y≤Y≤y+dy } =P{x≤X≤x+dx|y≤Y≤y+dy}

In other words, for small values of dx and dy,fX|Y(x|y)dx represents the conditional probability that X is between x and x+dx given that Y is between y and y+dy.

The use of conditional densities allows us to define conditional probabilities of events associated with one random variable when we are given the value of a second random variable. That is, if X and Y are jointly continuous, then, for any set A,

P{X∈A|Y=y}=∫AfX|Y(x|y) dx

In particular, by letting A=(−∞,a) we can define the conditional cumulative distribution function of *X* given that *Y* = *y* by

FX|Y(a|y)≡P{X≤a|Y=y}=∫−∞afX|Y(x|y) dx

The reader should note that by using the ideas presented in the preceding discussion, we have been able to give workable expressions for conditional probabilities, even though the event on which we are conditioning (namely, the event {Y=y}) has probability 0.

If X and Y are independent continuous random variables, the conditional density of X given that Y=y is just the unconditional density of X. This is so because, in the independent case,

fX|Y(x|y)=f(x,y)fY(y)=fX(x)fY(y)fY(y)=fX(x)

Example 5a

The joint density of X and Y is given by

f(x,y)={ 12 5 x(2−x−y) 0<x<1,0<y<1 0 otherwise

Compute the conditional density of X given that Y=y, where 0<y<1.

Solution

For 0<x<1,0<y<1, we have

f X|Y (x|y) = f(x,y) f Y (y) = f(x,y) ∫ −∞ ∞ f(x,y) dx = x(2−x−y) ∫ 0 1 x(2−x−y) dx = x(2−x−y) 2 3 −y/2 = 6x(2−x−y) 4−3y

Example 5b

Suppose that the joint density of X and Y is given by

f(x,y)={ e −x/y e −y y 0<x<∞,0<y<∞ 0 otherwise

Find P{X>1|Y=y}.

Solution

We first obtain the conditional density of X given that Y=y.

$$f_{X|Y}(x|y) = f(x,y)\, f_Y(y) = e^{-x/y} e^{-y} / y\, e^{-y} \int_0^\infty (1/y) e^{-x/y}\, dx = 1\, y\, e^{-x/y}$$

Hence,

$$P\{X>1|Y=y\} = \int_1^\infty 1\, y\, e^{-x/y}\, dx = -e^{-x/y} \mid_1^\infty = e^{-1/y}$$

Example 5c The t-distribution

If Z and Y are independent, with Z having a standard normal distribution and Y having a chi-squared distribution with n degrees of freedom, then the random variable T defined by

$$T = ZY/n = n\, ZY$$

is said to have a *t-distribution* with n degrees of freedom. As will be seen in **Section 7.8**, the t-distribution has important applications in statistical inference. At present, we will content ourselves with computing its density function. This will be accomplished by using the conditional density of T given Y to obtain the joint density function of T and Y, from which we will then obtain the marginal density of T. To begin, note that because of the independence of Z and Y, it follows that the conditional distribution of T given that Y=y is the distribution of n/y Z, which is normal with mean 0 and variance n/y. Hence, the conditional density of T given that Y=y is

$$f_{T|Y}(t|y) = \frac{1}{2\pi n/y}\, e^{-t^2 y/2n}, \quad -\infty<t<\infty$$

Using the preceding, along with the following formula for the chi-squared density given in **Example 3b** of this chapter,

$$f_Y(y) = e^{-y/2} y^{n/2-1} 2^{n/2}\, \Gamma(n/2), \quad y>0$$

we obtain that the joint density of T,Y is

$$f_{T,Y}(t,y) = \frac{1}{2\pi n}\, 2^{n/2}\, \Gamma(n/2)\, e^{-t^2 y/2n} e^{-y/2} y^{(n-1)/2} = \frac{1}{\pi n}\, 2^{(n+1)/2}\, \Gamma(n/2) e^{-t^2+n\, 2n}\, y^{(n-1)/2}, \quad y>0, -\infty<t<\infty$$

Letting $c=\frac{t^2+n}{2n}$, and integrating the preceding over all y, gives

$$f_T(t) = \int_0^\infty f_{T,Y}(t,y)\, dy = \frac{1}{\pi n}\, 2^{(n+1)/2}\, \Gamma(n/2) \int_0^\infty e^{-cy}\, y^{(n-1)/2}\, dy = c^{-(n+1)/2} \pi n\, 2^{(n+1)/2}\, \Gamma(n/2) \int_0^\infty e^{-x} x^{(n-1)/2}\, dx \quad \text{(by letting } x=cy) = n^{(n+1)/2}\, \Gamma(\tfrac{n+1}{2})\, \pi n\, (t^2+n)^{(n+1)/2}\, \Gamma(\tfrac{n}{2})$$

$$\left(\text{because } \frac{1}{c} = \frac{2n}{t^2+n}\right) = \frac{\Gamma(\frac{n+1}{2})}{\pi n\, \Gamma(\frac{n}{2})}\, (1+t^2/n)^{-(n+1)/2}, \quad -\infty<t<\infty$$

Example 5d The bivariate normal distribution

One of the most important joint distributions is the bivariate normal distribution. We say that the random variables X,Y have a bivariate normal distribution if, for constants $\mu_x, \mu_y, \sigma_x>0, \sigma_y>0, -1<\rho<1$, their joint density function is given, for all $-\infty<x,y<\infty$, by

$$f(x,y) = \frac{1}{2\pi\, \sigma_x \sigma_y \sqrt{1-\rho^2}}\, \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\, \frac{(x-\mu_x)(y-\mu_y)}{\sigma_x \sigma_y}\right]\right\}$$